Letter

Optical neuromorphic computing based on chaotic frequency combs in nonlinear microresonators

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In this work, we present an unique implementation of delay line-free reservoir computing based on state-of-the-art photonic technologies, which exploits chaotic optical frequency comb formation in an optical microresonator as the nonlinear reservoir. Our solution leverages the high resonator quality (Q)-factor both for memory and for enhancing high-dimensional nonlinear mapping of input symbols. We numerically demonstrate the accurate prediction of about one thousand symbols in chaotic time series without the need of dedicated optimization for specific tasks. Our results will enable design of optical neuromorphic computing architectures combining on-chip integrability, low footprint, high speed, and low power consumption.

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Optical reservoir computing (RC) is a computational approach that leverages potential benefits of photonic platforms, such as low latency, high bandwidth, fast reconfigurability, and low power consumption [1,2], to perform computation in a neuromorphic fashion going beyond the traditional von Neumann paradigm. RC is a neuromorphic computing approach within the recurrent neural network (RNN) family [3]. Its operational principle relies on a mapping of the input signal into a higher-dimensional space exploiting the nonlinear dynamics of a reservoir. By training only the output layer, without changing the inner layer's parameters, RC significantly reduces training time compared to standard RNN, making it an efficient and scalable solution for various computational tasks, including time-series prediction, pattern recognition, and nonlinear signal processing.

Photonic solutions for RC based on wave dynamics have been proposed [4–6] and have been implemented in several platforms, including optical fibers [7,8], optical parametric oscillators [9,10], saturable absorbers [11], media exhibiting quadratic nonlinearity [12], disordered scattering media [13], semiconductor optical amplifiers [14,15], semiconductor laser arrays [16], phase-change materials [17], and optical cavities and resonators [7,18–23]. Among photonic hardware for optical RC, microresonators (MRs) are promising due to compact dimensions, possible on-chip integrability, complementary metal-oxide semiconductor compatibility, and low power consumption through nonlinear response achievable with input power in the mW order, enabled by high-quality (Q)-factor platforms. In particular, so far, both theoretical and experimental investigations of RC in semiconductor MRs have been performed leveraging two-photon absorption and free carrier dispersion as nonlinear tools for achieving classification and prediction tasks [18–24]. Several approaches to optical RC use delay lines—for instance, in MR or in other settings as well—to enable system memory and symbol interaction [18,19,24,25]. However, in order to reduce system complexity and footprint, a particularly appealing emerging trend is the delay line-free reservoir computer [19,26], which does not require the use of auxiliary delay lines.

In this work, we present an unique method for optical RC without delay line, which exploits four-wave mixing-assisted chaotic optical frequency combs (OFCs) in high-Q-factor Kerr MRs. Our approach (1) proposes for the first time to exploit modulation instability (MI) gain-induced OFC generation to perform optical computing; (2) leverages intrinsic high-Q MR memory capabilities bypassing the need of adding extra delay lines, which enhances computational speed; and (3) enables nonlinear interaction between optical modes of the systems achievable for few tens of mW optical power injection to operate higher dimensionality transformation of input symbols necessary for computation, hence offering low power consumption. Optical frequency combs are ultraprecise optical rulers consisting of equally spaced coherent frequency laser lines that can usually be generated by several platforms, including mode-locked lasers [27], electro-optic modulators [28], and optical microresonators [29]. OFCs possess a variety of applications ranging from optical clocks to microscopy and from sensing to space exploration and telecommunications, to mention just a few [30,31]. While for the vast majority of applications it is required that high stability and a fixed phase relation exist between the comb modes, in this work we leverage the potential technological application of less explored but easier to generate OFCs exhibiting random amplitude and phase fluctuations in their spectral lines. The proposed RC scheme operates as follows: a symbol time series generated from a chaotic dynamical system is encoded, after suitable rescaling, into input pump power values at which an optical high-Q-factor Kerr MR is driven. The symbol input (driving

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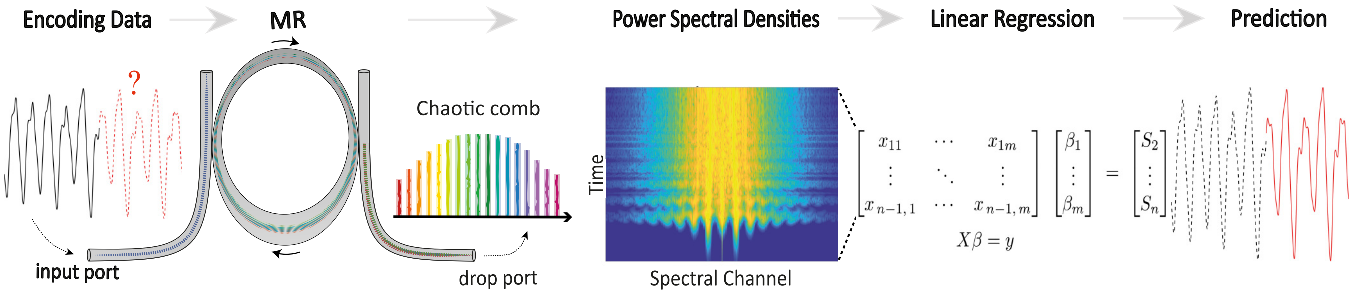

FIG. 1. Overview of the proposed RC model. The driving field encodes time-series information into the light wave, which drives the MR. The input light then enters the MR. The MR high-dimensional output state consists of power spectral densities corresponding to a chaotic frequency comb. Linear regression is then applied to combine generated spectral features, and then the next symbols in the chaotic time series are predicted.

pump power) is updated at a rate of 1 GSa/s. The nonlinear dynamics of light inside the resonator, due to the MI process, induces a dramatic spectral broadening leading to the generation of a chaotic frequency comb corresponding to a large number of oscillating equally spaced cavity resonances with amplitudes and phases that randomly change in time (see the Supplemental Material for more details [32]). The readout layer consists of the power spectral densities of the MR individual resonances as they can be detected by an optical spectrum analyzer. After training the reservoir, we applied linear regression to combine the output features and predict the next symbol in the time series. In high-Q resonators, the photon lifetime, given by $\tau_p = Q/\omega_0$, with ω_0 being the carrier frequency, can significantly exceed the inverse of the input symbol rate. This extended photon lifetime ensures that the spectral features generated by one input symbol persist in the cavity long enough to overlap with subsequent symbols, enabling interaction of different symbols inside the cavity. This is further enhanced by the bistable response of Kerr cavities, which is associated with a hysteresis cycle of the intracavity power as a function of the injection. Figure 1 provides an overview of the proposed RC model, illustrating each step: from data encoding till time-series prediction.

We modeled the dynamics of light inside the MR using the established mean-field Lugiato-Lefever equation in the notation of Ref. [33], which describes the evolution over time t of the electric field slowly varying envelope $E(t, \tau)$:

$$\frac{\partial E}{\partial t} = -\frac{(\alpha + i\delta_0)}{t_R}E - i\frac{L}{t_R}\frac{\beta_2}{2}\frac{\partial^2 E}{\partial \tau^2} + i\frac{\gamma L}{t_R}|E|^2 E + \frac{\sqrt{\theta}}{t_R}E_{\text{in}}. \tag{1}$$

Here, t_R represents the round-trip time of the cavity, δ_0 is the detuning of the driving field from the nearest linear resonance, L is the cavity length, β_2 is the group velocity dispersion coefficient, γ is the nonlinearity coefficient, and E_{in} is the continuous wave pump injection amplitude related to the injected power by $P_{\text{in}} = |E_{\text{in}}|^2$. α and θ characterize cavity losses and coupling strength as a function of the resonator Q factor, which is defined as $Q = (Q_{\text{int}}^{-1} + Q_{\text{ext1}}^{-1} + Q_{\text{ext2}}^{-1})^{-1}$ and depends on the contributions from internal losses Q_{int} and losses associated with the input, Q_{ext1}, and drop, Q_{ext2}, waveguides. To numerically solve Eq. (1), we employed a fourth-order Runge-Kutta interaction picture method with local tolerance control (see the Supplemental Material for more details about parameters definitions and numerical methods [32]).

We tested the prediction capabilities of our MR-based optical reservoir computer on three different tasks, namely, in the prediction of chaotic time series generated by different dynamical systems: Mackey-Glass, Rössler, and Lorenz equations. We run simulations of these dynamical systems (see the Supplemental Material for more details [32]), generating for each one a time series featuring 1700 symbols. We then mapped the time series into pump power values $P_{\text{in}} \in [P_{\text{av}} \pm \Delta/2]$, where P_{av} is the average power and Δ defines the pump power variation, e.g., the excursion between maximum and minimum pump power injected. Such mapping could be experimentally implemented by temporally controlling the power of the continuous wave pump laser by means of a modulator. For each value of the resonator input pump power, we built the RC feature matrix having as rows the power spectral densities collected from the MR drop port at regular time intervals—in the nanosecond scale (see individual picture captions)—corresponding to the inverse of input symbol rate. The power spectral densities (shown in the "Power Spectral Densities" section of Fig. 1), representing the high-dimensional states generated by the MR, are hence transformed into features that form the input to the linear regression model. Each row of the feature matrix corresponds to a specific input symbol, and the matrix is then used to predict the next symbol in the time series. To enhance stability and emphasize relevant features, the power spectral density sequence is first filtered around the pump frequency using a super-Gaussian filter of order 4 and a width of 1 THz. For each series, we used 80% of the data points for training and 20% for testing, resulting in a feature matrix of 1360×512 with a rank typically on the order of 512. We then applied linear regression to estimate the optimal weight array $(\beta_1, \ldots, \beta_m)$ (see the Supplemental Material [32]). When constructing the feature matrix, we also took into account the initial phase of intracavity energy buildup. This transient "warm-up" period was removed from the feature matrix to ensure that only quasi-steady-state characteristics of the system were included. The performance of the RC system on each task was evaluated using a standard indicator, namely, the normalized mean squared error (NMSE), which quantifies the prediction error (see the Supplemental Material for the NMSE definition [32]). Figure 2 illustrates the single-step

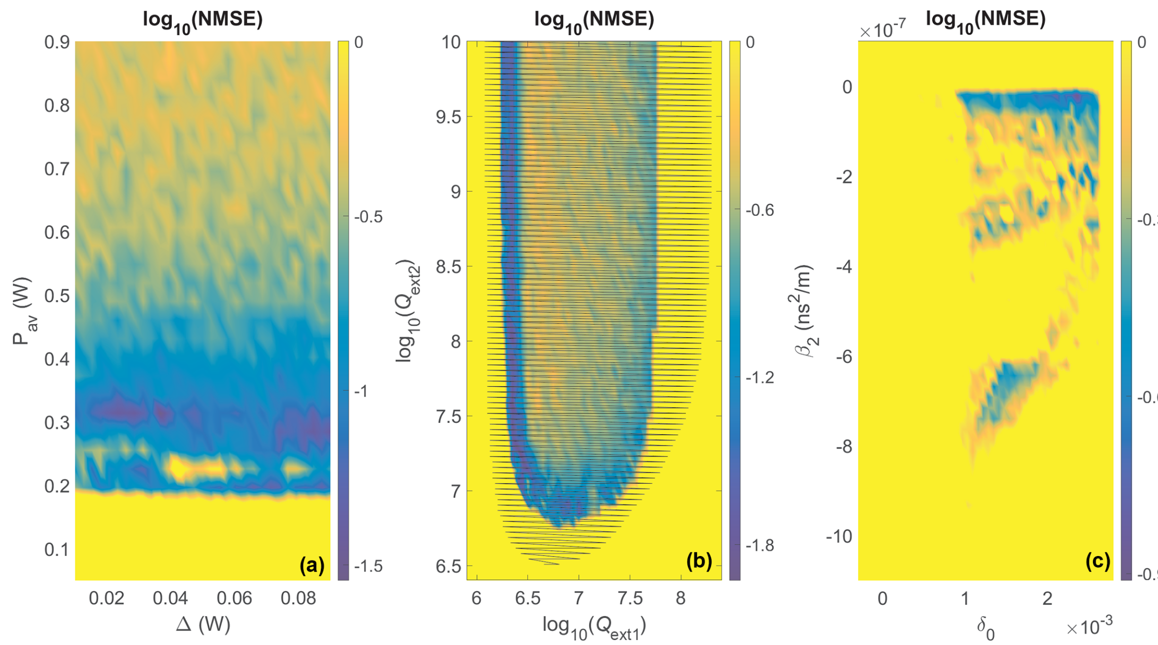

FIG. 2. (Mackey-Glass) Single-step prediction performance. (a) $\log_{10}$(NMSE) as a function of average pump power (P_{av}) and pump power variation (Δ). (b) $\log_{10}$(NMSE) as a function of external quality factors Q_{ext1} and Q_{ext2}. The shaded area shows the region where MI gain is larger than zero. (c) $\log_{10}$(NMSE) as a function of β_2 versus δ_0. The color scale indicates the NMSE, with lower values representing better prediction performance. In panels (a) and (c), $Q_{\mathrm{ext1}} = Q_{\mathrm{ext2}} = Q_{\mathrm{int}} = 10^7$ and total Q factor $= 3.33 \times 10^6$, and in panel (b), $Q_{\mathrm{int}} = 10^7$. Common parameters are $\gamma = 1.6 \times 10^{-2}\ \mathrm{W}^{-1}\,\mathrm{m}^{-1}$, $\beta_2 = -4.0 \times 10^{-9}\ \mathrm{ns}^2\,\mathrm{m}^{-1}$, $\delta_0 = 1.6 \times 10^{-3}$, $\Delta = 0.05\ \mathrm{W}$, $P_{\mathrm{av}} = 0.325\ \mathrm{W}$, and the data rate is 1.0 GSa/s.

prediction performance as a function of the Q factor, detuning δ_0, and dispersion β_2. The color scale represents the prediction error, with lower NMSE values indicating better prediction performance. The MI process requires a minimum amount of average power to start, which translates into a clear threshold, $P_{\mathrm{av}} \approx 0.2\ \mathrm{W}$, needed to achieve low NMSE value as we can appreciate from Fig. 2(a). Figure 2(b) shows the computer performance as a function of external Q factors (Q_{ext1} and Q_{ext2}). We observe that the reservoir computer is capable of prediction [$\log_{10}(\mathrm{NMSE}) < 0$] when MI gain is different from zero. Hence, MI is a necessary condition for the operation of our device. From Fig. 2(b), we can also appreciate the existence of a region of optimal performance [dark blue stripe for $\log_{10}(Q_{\mathrm{ext1}}) \approx 6.5$]. We furthermore verified that this area moves to the left (lower values of Q_{ext1} are needed to achieve optimal performance) when decreasing the input symbol rate. For input data rate slower than $\approx$0.1 GSa/s and for input data rate on the order of 10 GSa/s, the device is not successful in performing prediction (see the Supplemental Material for more details [32]). We can intuitively explain these limitations as follows: when input data rate is too slow, the currently injected symbol dominates the dynamics and the memory of previous symbols is lost; on the other hand, when input data rate is very high, the new symbols do not have enough time to build up and to leave their "imprint" inside the cavity. Hence, an optimal input data rate exists, which reaches a trade-off between these two effects. In Fig. 2(c), $\log_{10}$(NMSE) is shown as a function of group velocity dispersion β_2 and detuning δ_0. We can qualitatively explain optimal performance of the system [blue areas corresponding to low NMSE in Fig. 2(c)] as for low absolute values of anomalous dispersion the parametric gain spectrum is broader, which enables the generation of a larger number of spectral features. Narrow range of detuning values for which the performance is optimal can be interpreted as follows. For $\delta_0 \approx 0$, the cavity is monostable. By increasing the value of δ_0 around 10^{-3}, much larger average intracavity power can be reached due to the appearance of bistability in the cavity response: with relatively low injected pump power on the order of few hundreds of mW, the intracavity power can be in excess of 100 W, which can lead to broadband parametric gain and random OFC formation. By increasing even further the value of the detuning—$\delta_0 > 2.5 \times 10^{-3}$—the upper branch of the cavity response is bistable for values of the injection much larger than those used in our simulations—$P_{\mathrm{in}} \approx 0.3\ \mathrm{W}$—hence, only the lower branch of the bistability curve can be accessed (see the Supplemental Material [32]). It is also important to stress that existence of MI is a necessary condition for the operation of our optical reservoir computer, but it does not explain alone the parameter regions of optimal performance [dark blue regions in Fig. 2(b)]. We have performed the same characterization for time series generated from the Rössler and Lorenz systems, and the results were similar to those obtained for the Mackey-Glass time series as detailed in the Supplemental Material [32].

The impact of input and readout signal-to-noise ratio (SNR) on prediction accuracy—that experimentally can be due to imperfection at the encoding or detection stage—has been analyzed too: Our system can achieve good performance for input and readout SNR higher than 20 dB (see the Supplemental Material [32]). We have tested our RC in more advanced multistep prediction tasks. We began with the same

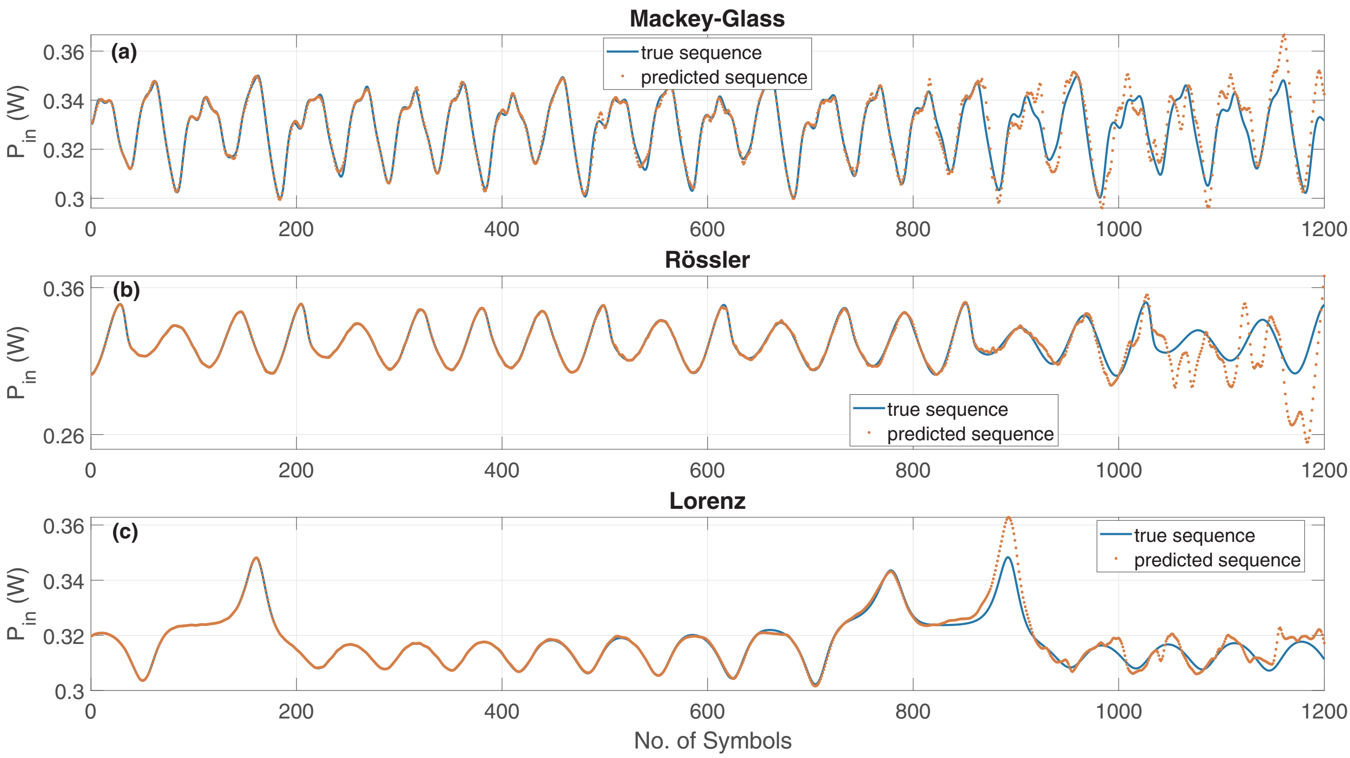

FIG. 3. Examples of multistep predictions using features generated from a single symbol for three benchmark dynamical systems: (a) Mackey-Glass time series, (b) Rössler attractor, and (c) Lorenz system. Each subplot shows the true sequence (blue line) alongside the predicted sequence (red dots) over 1200 symbols. Parameters are the same as in Fig. 2.

procedure as in single-step prediction, where the initial internal state of the MR is used to generate spectral features from the first symbol. These features are then used in a pseudoinverse linear regression model to predict the next symbol (input power). The predicted power is fed back into the MR to generate new features, continuing the cycle in an autoregressive manner. This process allows the system to predict subsequent symbols, achieving predictions up to 1200 symbols ahead of a single input symbol. Figure 3 illustrates examples of multistep prediction using features generated from a single symbol for the Mackey-Glass time series [Fig. 3(a)], Rössler attractor [Fig. 3(b)], and Lorenz system [Fig. 3(c)]. The figure displays the true sequences (blue line) and the predicted sequences (red dots), showing that the predicted power values closely follow the true patterns. The regression method used is pseudoinverse, with 1700 symbols for training and testing and 700 symbols for warm-up.

In order to quantify the multistep prediction capability of our RC, we calculated the RC error $\epsilon = |y_{\text{true}} - y_{\text{pred}}|$, where y_{true}, y_{pred} stand for the true and predicted values, respectively, for the multistep predictions across 20 time series, each obtained by seeding the dynamical systems with different input conditions. Results are reported in Fig. 4.

Our chaotic comb-based RC provides a solution for fast optical RC with excellent computational performance and a compact setup that exploits state-of-the-art integrated photonic devices. One of its advantages, compared to many alternative MR-based architectures [18,23,24], is the delay-free approach, which enables a compact architecture while also taking advantage of the high intrinsic resonator factor Q for memory functionality. The potential computational speed of our device is in the GHz range, exceeding a few hundred MHz reported in other MR-based RC computing approaches [19]. Moreover, in our system, photon lifetime, which is associated with memory, is directly influenced by the total Q factor. For the range of Q factors with low NMSE, the calculated photon lifetime ranges between 1.78 and 5.62 ns. Since the lowest photon lifetime exceeds the 1 ns sampling rate, this ensures that at least two symbols interact within the cavity, enabling effective memory retention and improving performance in tasks like time-series prediction. Further analysis of the system's memory capacity [34,35], detailed in the Supplemental Material [32], confirms that the memory retention is maximized at an optimal symbol rate of 1 GSa/s, where the system achieves a peak memory capacity of 8. This is also the optimal data rate for prediction performance, reflecting a correlated trend between the system's memory capacity and its predictive capability. Moreover, exploring the impact of detuning shows that this device can be reconfigured to independently maximize either memory or computing power (see the Supplemental Material [32]). Silicon MR RC schemes based on two-photon absorption and free carrier dispersion effects can operate at input power on the order of few mW due to the low threshold of the nonlinear dynamics exploited; however, they have not demonstrated multiple symbol prediction capability to the best of our knowledge [18–24]. While we have reported RC operation in the range of 0.2–0.3 W input pump power, it is worth stressing that in our simulations we have taken a conservative approach choosing a relatively small value of Kerr nonlinearity

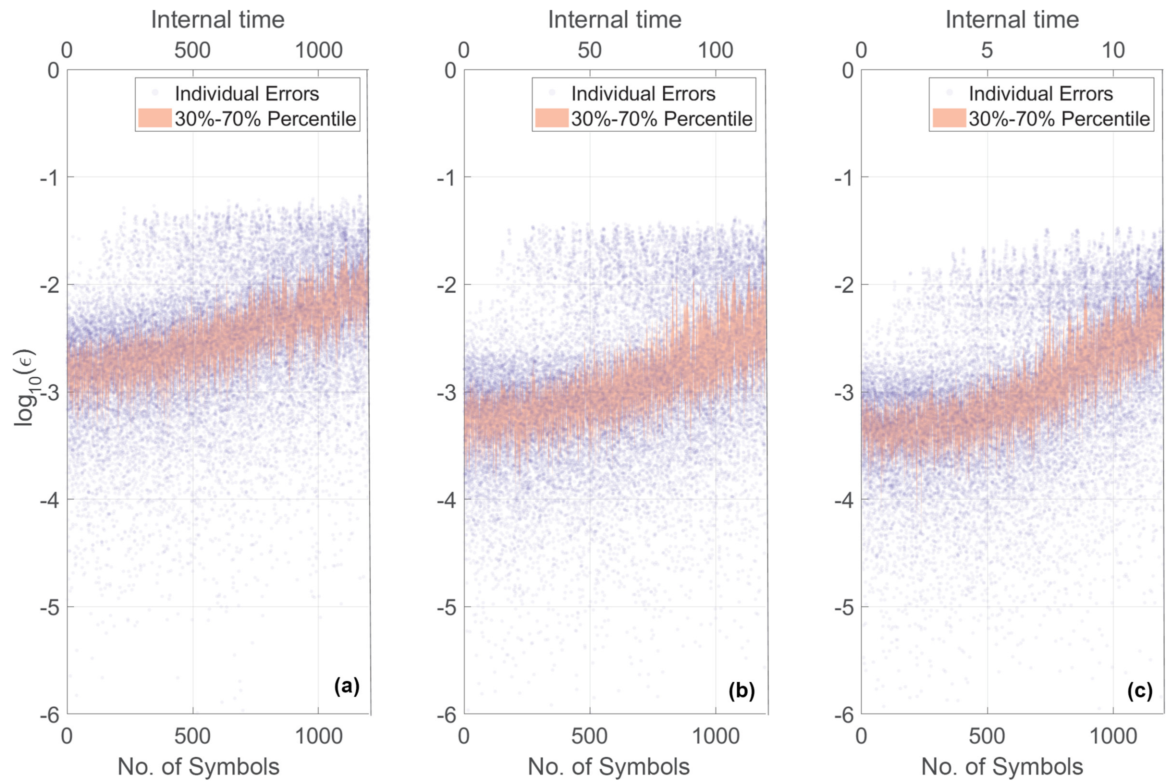

FIG. 4. Error analysis of multistep autoregressive predictions for three benchmark dynamical systems: (a) Mackey-Glass time series, (b) Rössler attractor, and (c) Lorenz system. The plot shows the logarithmic scale of the absolute prediction errors, $\log_{10}(\epsilon)$, for each of the 1200 symbols in the 20 predicted sequences. Individual errors are represented as purple dots, while the shaded pink region indicates the 30th to 70th percentile range of the errors, demonstrating the spread around the median. The lower x axis, labeled as "No. of Symbols" represents the "depth" of the prediction, with each of the predicted symbols consecutively being encoded in the amplitude of light and transmitted to the MR. The upper x axis, "Internal time," represents the corresponding timescale of evolution of the dynamical system that we use to test our RC. Parameters used are the same as in Fig. 3.

coefficient γ compatible with fused silica MRs. Much larger values—more than two orders of magnitude—are used in the literature for MR made of materials such as silicon nitride, or magnesium fluoride [33], or others [36]. For larger values of γ, the MI process triggering spectral broadening can occur at lower input power. In our work, we have also presented an application of chaotic OFCs having nonlocked phases of neighboring cavity modes. Chaotic comb states have so far mostly been avoided for applications due to their intrinsic instability with limited exceptions in the field of light detection and ranging (LIDARs) [37,38]. To conclude, we have proposed a method for delay-free optical reservoir computing based on chaotic frequency combs spontaneously generated by nonlinear four-wave mixing dynamics in a driven optical Kerr microresonator, demonstrating multistep prediction of about 1000 consecutive symbols in chaotic time series with task-independent operation. Our results open new research directions for low power consumption and low-footprint optical reservoir computing on a chip with ultrafast optical components.

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